

The Dollar's Safe-Haven Status Across Two Shocks: Evidence from Liberation Day and the Strait of Hormuz Crisis

Bachelor's Thesis

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1 Introduction

Within a single year, two major shocks moved global commodity and financial markets. On the 2nd of April 2025, widely referred to as "Liberation Day", the United States Government announced broad-based tariffs on virtually all of its trading partners, citing persistent trade deficits and national security concerns. The policy included a base 10% tariff and, subsequently, additional higher rates for trading partners with large surpluses against the U.S., such as China and the European Union. Following the announcement, the dollar fell sharply against a broad basket of currencies, with the Euro, Yen and Swiss Franc appreciating by 3–4% in the first week alone. The move was accompanied by a spike in equity-market volatility and a U.S. equities sell-off.

Less than a year later, on 28 February 2026, the United States and Israel launched airstrikes on Iran, beginning the 2026 Iran war. As escalation in the Gulf threatened to close the Strait of Hormuz and sent crude sharply higher, the dollar did exactly what the safe-haven literature predicts—it appreciated. Two episodes, both with the United States implicated, produced a mirror image. This thesis asks why.

The puzzle speaks to a live debate about whether the dollar's exceptional status is eroding. A growing literature argues that the currency's safe-haven premium has weakened, pointing to the unusual co-movement of a falling dollar, rising Treasury yields and rising risk premia during the 2025 tariff episode (Ostry et al. 2025; Kamin 2025; Jiang et al. 2026). Against this, the long tradition on the dollar's "exorbitant privilege" and its role as the world's reserve and safe-haven currency (Gourinchas and Rey 2005; Obstfeld and Zhou 2023; Habib and Stracca 2012; Ranaldo and Söderlind 2010) implies the status should be durable. The two 2025–2026 shocks offer an unusually clean natural experiment for adjudicating the debate, because they differ in *kind*: one is a U.S. trade-policy shock, the other a geopolitical oil-supply shock.

The central question of this thesis is therefore: *does the dollar's safe-haven response depend on the type of shock—even when the United States is involved in both?* To adjudicate it, the design sets the two shocks against a historical benchmark—three episodes in all. The Liberation Day tariff announcement is a purely U.S.-originated economic-policy shock. The 2026 Israel–Iran/Hormuz crisis is a geopolitical oil-supply shock in which the United States is implicated as a co-belligerent but still functions as a financial safe harbour. And the 2022 Russian invasion of Ukraine—an external military/oil shock not authored by the United States—serves as a control for the "classical" safe-haven response. Comparing the three isolates whether the dollar's

1 Introduction

behaviour is conditioned by the nature of the shock or merely by who caused it.

The thesis answers the question with two complementary empirical layers on a single daily dataset spanning January 2019 to June 2026. An *event study* measures the cumulative abnormal response of the dollar, a cross-section of currency baskets, gold, equities and crude oil around each shock, and tests formally whether the dollar’s response differs across events. A complementary *regression* then tests whether the dollar’s relationship with risk itself changed across the episodes—whether its safe-haven sensitivity to volatility held or broke. The two layers are complementary: the event study isolates the dollar’s move around each discrete, dated shock, while the regression tests whether its underlying sensitivity to risk shifted across the episodes.

The contribution is threefold. First, to the best of the author’s knowledge this is the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets. Second, the thesis bridges the safe-haven and commodity-currency literatures by carrying the oil channel and a gold benchmark through the same design, so that the dollar’s behaviour can be read against gold and the oil-exposed currency cross-section rather than in isolation. Third, the rebuilt carry/dollar risk factors and the WMR currency panel extend standard cross-sectional tools through to mid-2026, allowing the most recent shocks to be studied with consistent, citable data.

The remainder of the thesis is organised as follows. Chapter 2 reviews the safe-haven, dollar-erosion and commodity-currency literatures. Chapter 3 sets out the conceptual framework—the working definition of a safe-haven currency, a taxonomy that separates a shock’s originator from its nature, and the hypotheses that follow. Chapter 4 narrates the 2022–2026 events. Chapter 5 describes the data and the event-study and regression methodologies. Chapter 6 presents the empirical results. Chapter 7 discusses the findings and their implications, and Chapter 8 concludes.

The results point to a dollar whose safe-haven status is conditional on the type of shock rather than on its originator. The two U.S.-involved shocks produce a mirror image: the Liberation Day tariffs depreciate the broad dollar index by over three and a half percent within twenty trading days, while the Hormuz oil-supply crisis appreciates it—by as much as the external Ukraine benchmark, from which its response is statistically indistinguishable. The complementary regression confirms the pattern: the dollar’s risk sensitivity, significantly positive on normal days, collapses to essentially zero inside the tariff window yet holds through the Hormuz and Ukraine episodes. Gold emerges as a hedge against the dollar rather than a universal safe haven, and the oil channel sorts the currency cross-section only for the supply-driven

shock.

2 Literature Review

This chapter situates the thesis within six distinct strands of literature. Each body of research addresses a specific dimension of our core question: does the dollar’s status as a safe haven depend on the nature of a shock, rather than its source? By reviewing these fields, this chapter establishes the theoretical framework to understand how the dollar’s safe-haven functions, setting the stage for examining the consequences of when the United States itself originates a shock or acts as a co-belligerent.

2.1 Currency Risk Factors

[Lustig et al. \(2011\)](#) establish the foundational framework for understanding currency risk by identifying two primary common risk factors: a “level” factor, representing the dollar’s average fluctuations against all foreign currencies, and a “slope” factor, which captures the excess returns of carry trades. They demonstrate that high-interest-rate currencies load more heavily on the slope factor, which is closely linked to global equity-market volatility. Crucially, their model shows that during periods of high global risk, low-interest-rate currencies endogenously appreciate, acting as a hedge or safe haven, while high-interest-rate currencies depreciate. This suggests that the dollar’s behavior as a safe haven is intrinsically tied to its role as a low-yield currency that provides insurance against global risk by appreciating during adverse shocks.

Furthermore, the dollar’s safe-haven role is amplified when shocks interact with specific trade and financial frictions. [Dalgic and Ozhan \(2025\)](#) argue that the dollar’s response is endogenously determined by an economy’s level of dollar invoicing and dollar-denominated debt. In their framework, a shock—such as a foreign interest-rate hike—that triggers a depreciation in a small open economy becomes recessionary because it inflates the real burden of dollar debt and mutes expenditure switching. This countercyclical co-movement between GDP and the exchange rate renders the local currency a poor hedge, forcing a flight to the safety of the dollar. For the empirical design, this strand supplies the building blocks: the dollar and carry factors rebuilt through mid-2026 underpin the portfolio construction of Chapter 5.

2.2 Dollar Dominance and Its Erosion

The dollar’s role as the primary global reserve currency confers an “exorbitant privilege”—a persistent excess return on U.S. external assets over its liabilities

(Gourinchas and Rey 2005)—mirrored by an “exorbitant duty”: in global crises the United States transfers wealth to the rest of the world, roughly 19% of U.S. GDP during 2007–2009, as the dollar’s flight-to-safety appreciation revalues its external balance sheet (Gourinchas et al. 2017). On the mechanism, Jiang et al. (2021) identify surges in foreign demand for safe dollar assets—“safety shocks”—that widen the Treasury basis and appreciate the dollar immediately: precisely the reflex a safe-haven currency should display.

Crucially, this literature implies that the haven response depends on what a shock does to the demand for dollar safety. While the 2008 financial crisis and the COVID-19 pandemic elicited the standard appreciation, Jiang et al. (2026) document an anomalous depreciation after the 2025 U.S. tariff announcements—a “flight away” from the dollar examined in detail in Section 2.5. Obstfeld and Zhou (2023) add the cyclical dimension: a “global dollar cycle” in which dollar appreciation tightens financial conditions worldwide, with dollar movements correlating most reliably with investor risk appetite. A slow-moving counterpart to any abrupt break is visible in official reserves, which Arslanalp et al. (2022) show have been actively diversified away from the dollar for two decades. For the thesis, this strand establishes the stakes: if the dollar’s haven status is structural, it should hold across shock types; if it is eroding, U.S.-originated policy shocks are where the cracks should show first.

2.3 Safe-Haven Currencies and Gold

The literature on safe-haven currencies indicates that the U.S. dollar’s status is not static but varies with the nature and epicentre of a shock. A central paradox, identified by Habib and Stracca (2012), is the dollar’s appreciation during the 2008 global financial crisis—an episode the United States itself exported. They find that while the net foreign asset position is the most robust long-term predictor of safe-haven status, the dollar’s 2008 behaviour was anomalous compared with historical regularities.

The comparative efficacy of the dollar across shocks is most directly addressed by Cheema et al. (2022), who find that the dollar’s safe-haven character weakened considerably during the COVID-19 pandemic compared with 2008: an intermediate haven in the earlier crisis, it performed weakly across most settings in the later one—the empirical counterpart of the demand-side reading of the same episodes in Section 2.2. Providing broader context, Bock and Carvalho Filho (2015) demonstrate that the dollar, Japanese yen and Swiss franc consistently appreciate during risk-off

episodes, with magnitudes shaped by pre-crisis yields and market-liquidity factors. At the highest frequencies, however, the ordering within pairs reverses: [Ranaldo and Söderlind \(2010\)](#) show that in U.S. equity sell-offs the franc and yen appreciate *against the dollar*, with these haven properties materialising within hours to days and strengthening non-linearly in volatility—so against the classic havens themselves, the dollar is the risky leg. Gold supplies the natural non-sovereign benchmark: [Baur and Lucey \(2010\)](#) find that gold is a hedge against equities on average but a safe haven only in extreme declines, and only briefly. This conditional, crisis-contingent notion of safety is the one the thesis adopts (Chapter 3) and tests for the dollar, with the yen, the franc and gold as the benchmarks against which its response is judged.

2.4 Geopolitical Risk

Recent research suggests that the dollar’s response to geopolitical shocks is contingent on the nature of the event and a country’s position in the global network. [Caldara and Iacoviello \(2022\)](#) provide the foundational measurement, distinguishing geopolitical “threats”—the expectation of future adverse events—from geopolitical “acts”, their actual realisation; a shock to their news-based index induces persistent declines in investment, employment and stock prices, driven by both the threat and the realisation components. Expanding this to exchange rates, [Yilmazkuday \(2025\)](#) demonstrates that the dollar’s geopolitical safe-haven status is not universal: shocks to global geopolitical risk—driven primarily by geopolitical acts—produce a statistically significant depreciation of the U.S. dollar, even as currencies such as the South African rand or Brazilian real appreciate, with the currencies of countries more deeply embedded in global value chains depreciating most. Complementing this, [Liu and Zhang \(2024\)](#) find that a strategy long high-GPR currencies and short low-GPR currencies earns a significant excess return, and that this return predictability stems from country-specific and regional risk components rather than a single global factor—implying that a currency’s risk–return profile is sensitive to the geographic origin of the shock. The thesis draws on this strand directly: the [Caldara and Iacoviello \(2022\)](#) index dates and scales the Hormuz and Ukraine episodes, its threat–act distinction maps onto the staged escalation of the Hormuz crisis, and the GPR series serves as an external control in the empirical chapters.

2.5 Tariff Shocks and Exchange Rates

The 2025 tariff episode has generated a fast-growing literature that is central to this thesis. [Kamin \(2025\)](#) provides econometric evidence that, following the Liberation Day announcements of April 2025, the dollar switched from behaving as a safe haven—appreciating in periods of market volatility—to behaving as a “risk-on” currency that moves inversely with volatility. [Jiang et al. \(2026\)](#) document the same break in historical correlations—the dollar depreciating while U.S. rates rose relative to the euro, the VIX increased, and the dollar convenience yield fell—and interpret it through shifts in global demand for dollar safe assets, calibrating a steady-state real dollar depreciation of around 7.6% should the United States lose reserve-currency demand. [Hassan et al. \(2025\)](#) provide the theoretical counterpart: in a general-equilibrium model in which currency safety is endogenous, a trade war that isolates U.S. goods markets erodes the dollar’s safety premium and raises U.S. interest rates.

On the empirical exchange-rate response, [Ostry et al. \(2025\)](#) construct a tariff-shock database and show that exchange rates react to U.S. tariff shocks systematically once the size and relevance of the measures—and expected retaliation—are accounted for. Notably, the 2025 depreciation inverts the pre-2025 evidence, in which U.S. tariff actions tended to *strengthen* the dollar: [Khalil and Strobel \(2024\)](#) trace the dollar’s appreciation during the 2018–2019 tariffs to trade policy itself, making the 2025 episode a genuine puzzle. Closest to the present design, [Benguria and Saffie \(2025\)](#) run an event study of the 2 April 2025 announcement and identify sizeable declines in U.S. equity prices that scale with the tariff increment. This strand supplies the thesis’s first treatment: the Liberation Day shock, whose event-study comparison against contemporaneous military/oil-supply shocks is the contribution of this thesis.

2.6 Commodity Currencies and Oil

[Y.-C. Chen and Rogoff \(2003\)](#) establish that the U.S.-dollar price of commodity exports exerts a strong and stable influence on the real exchange rates of commodity exporters such as Australia, Canada and New Zealand, providing the basis for the oil-exporter versus oil-importer portfolio split used in Chapter 5. The nature of the oil shock matters, however: [Kilian \(2009\)](#) shows that oil-price movements driven by supply disruptions, aggregate demand, and precautionary (oil-specific) demand have distinct macroeconomic consequences, so a price increase caused by a supply

scare is not equivalent to one caused by strong demand—nor to a tariff-induced demand collapse. This typology maps directly onto the thesis’s contrast between the demand-side tariff shock (falling oil) and the supply-side Hormuz shock (surging oil). The oil–dollar link also runs through the exchange rate directly: [H. Chen et al. \(2016\)](#) show that the dollar’s response to oil-price shocks depends on whether they originate in supply or demand. The anatomy of the 2026 Hormuz episode itself, and the sources documenting it, are set out in Chapter 4.

2.7 Synthesis and Research Gap

Across these six strands, the dollar’s safe-haven status has been established structurally, as a feature of the international monetary system; conditionally, as a crisis-contingent property in the sense of [Baur and Lucey \(2010\)](#); and, most recently, as something the 2025 trade-policy shock may have disturbed. What the literature has not yet done is hold a U.S.-originated trade-policy shock against a contemporaneous military and oil-supply shock within a single research design, carrying the oil channel and a gold benchmark through the same lens. That comparison is the gap this thesis addresses. The next chapter distils these strands into a working definition of a safe-haven currency, a taxonomy that separates a shock’s originator from its nature, and the hypotheses that follow.

3 Conceptual Framework

This chapter explains the concepts the empirical analysis rests on. It states the working definition of a safe-haven currency, sets out a taxonomy that separates a shock’s *originator* from its *nature*, and distils the literature of Chapter 2 into the hypotheses the rest of the thesis tests.

3.1 Safe Haven, Hedge, and the Dollar’s Conditional Status

Following [Baur and Lucey \(2010\)](#), this thesis adopts a conditional definition. An asset is a *hedge* if it is uncorrelated with risky assets on average, and a *safe haven* if it is uncorrelated with – or negatively correlated with – them *specifically during market stress*. The distinction matters because safe-haven status is a crisis-contingent property rather than an average one: an asset can hedge in calm markets yet fail in a particular episode, or carry a haven bid only when risk spikes. Applied to currencies, the empirical signature of safe-haven behaviour is that the currency appreciates when global risk rises – when equities fall and volatility increases ([Ranaldo and Söderlind 2010](#); [Bock and Carvalho Filho 2015](#)). This is the property the thesis tests for the dollar, episode by episode, in the event study and the regression of Chapter 6.

The dollar’s claim to that status is structural. The literature on the “exorbitant privilege” ([Gourinchas and Rey 2005](#)) and on the global dollar cycle ([Obstfeld and Zhou 2023](#)) casts the dollar as the anchor of the international monetary system, the currency to which capital flees in a crisis. [Habib and Stracca \(2012\)](#) show that this status rests on fundamentals – a strong net foreign asset position, low external vulnerability, and economic size – rather than on the interest-rate spread against the United States.

3.2 A Taxonomy of Shocks

The thesis separates two attributes of a shock that empirical work often confounds: who *originates* it and what *kind* of shock it is. The originator may be the United States or an external actor; the nature may be an economic-policy shock – here, trade policy – or a military/oil-supply shock. Crossing these two dimensions, the three episodes studied occupy three distinct cells. The 2025 Liberation Day tariffs

are a U.S.-originated economic-policy shock. The 2026 Israel–Iran/Hormuz crisis is a military/oil-supply shock in which the United States is implicated as a co-belligerent. The 2022 Russian invasion of Ukraine is an external military oil-supply shock that the United States did not author.

This design is what allows origin and nature to be told apart. If the dollar’s response were governed by *origin*, the two U.S.-originated shocks – tariffs and Hormuz – would move the dollar alike and differ from the external Ukraine benchmark. If instead it were governed by *nature*, the two military/oil-supply shocks – Hormuz and Ukraine – would look alike regardless of U.S. involvement, and the trade-policy shock would stand apart. The two readings make opposite predictions about which episodes cluster together, and the empirical chapter adjudicates between them.

The nature of the oil move matters in its own right. [Kilian \(2009\)](#) distinguishes oil-price changes driven by supply disruptions from those driven by demand, with different macroeconomic consequences. The taxonomy maps onto that distinction: the Hormuz episode is a supply-driven oil shock – crude rises as a chokepoint is threatened – whereas the tariff shock depresses oil through weaker expected *demand*. The sign of the oil move therefore helps classify the shock and, through the commodity-currency channel ([Y.-C. Chen and Rogoff 2003](#)), sorts the currency cross-section between oil exporters and importers.

3.3 Hypotheses

Taken together, these strands point to a single organising prediction: the dollar’s safe-haven response is conditional on the *type* of shock rather than on whether the United States is its originator. Both the external benchmark (the 2022 Russian invasion of Ukraine) and the U.S.-originated Hormuz crisis are military/oil-supply shocks, around which the safe-haven literature implies the dollar should appreciate; the Liberation Day tariff announcement, by contrast, is a U.S. trade-policy shock that the dollar-erosion literature suggests should weaken the very currency it originates from. Gold offers a natural counterpoint: as an asset external to any sovereign it might be expected to bid in *both* kinds of episode, so that comparing it against the dollar isolates what is specific to the dollar’s status ([Baur and Lucey 2010](#)) – a prediction this thesis puts directly to the data. The cross-section of currencies should in turn sort on the oil channel, exporter currencies outperforming importers when the Hormuz supply shock lifts crude but losing ground when the tariff shock depresses oil through weaker demand ([Y.-C. Chen and Rogoff 2003](#); [Kilian 2009](#)).

4 Background: The Events of 2022–2026

This chapter establishes the institutional setting of the episodes the empirical chapters analyse and fixes the event dates that define the windows used in Chapter 5. For each episode it documents what happened, how official institutions characterised the market backdrop, and how the shock is classified under the originator–nature taxonomy of Section 3.2; each episode section closes with the response the safe-haven hypothesis of Section 3.3 predicts *ex ante*, so that Chapter 6 reads as a test of stated predictions rather than a search for patterns. The scholarly debate these episodes feed into is reviewed in Chapter 2 and is not repeated here.

4.1 Liberation Day and the U.S. Tariff Shock

On 2 April 2025, in an announcement branded “Liberation Day,” the United States imposed broad “reciprocal” tariffs on nearly all of its trading partners, combining a flat baseline rate with higher country-specific surcharges ([Kamin 2025](#)). The measure represented an abrupt change in the U.S. trade-policy regime rather than a marginal adjustment, and official assessments subsequently documented the resulting trade fragmentation and the pass-through of higher tariffs to economic activity ([European Central Bank 2025](#); [European Central Bank 2026](#)).

The market reaction was one of broad risk aversion: equities sold off, implied volatility rose, and crude oil fell as market participants priced weaker global demand ([European Central Bank 2025](#)). Around the announcements, the dollar depreciated against the euro, yen and Swiss franc even as risk metrics deteriorated, and long-dated Treasury yields rose while the currency weakened ([Kamin 2025](#)). The competing interpretations of this break—a repricing of the dollar’s reserve status and an erosion of its safety premium—are reviewed in Section 2.5.

Under the taxonomy of Section 3.2, Liberation Day is the study’s clearest U.S.-authored economic-policy shock, and the accompanying fall in crude marks it as a demand-side disturbance. A partial ninety-day tariff pause on 9 April 2025 tempered the equity drawdown without restoring the dollar ([Kamin 2025](#)), and is treated here as a separately dated reversal. Under the safe-haven hypothesis of Section 3.3, a U.S.-authored shock that erodes the dollar’s safety premium should weaken the currency on announcement and recover, at most, only partially when the policy is partially walked back.

4.2 The Israel–Iran Escalation and the Strait of Hormuz

The Gulf episode unfolded in stages, from threat to realisation. A first, twelve-day confrontation in June 2025 – Israeli and U.S. strikes on Iranian facilities, an Iranian response, and a rapid ceasefire – lifted crude only briefly and left major currencies little changed (Kilian et al. 2026). Escalation resumed in early 2026 and turned acute on 28 February 2026, when the United States and Israel launched airstrikes on Iran. Iran retaliated and moved to close the Strait of Hormuz – the maritime chokepoint through which roughly a fifth of globally traded oil passes – declaring it closed on 2 March and claiming complete control on 4 March 2026 (Fattouh and Economou 2026). Attacks on shipping followed and Brent crude rose sharply, trading above \$100 per barrel at the height of the disruption (International Monetary Fund 2026), with Kilian et al. (2026) tracing the inflationary consequences through the oil channel.

Under the taxonomy of Section 3.2, the crisis is a military/oil-supply shock in which the United States is involved as a co-belligerent, and the rise in crude identifies it as a supply-driven disturbance – the mirror image of the demand-side tariff shock. Several further dated events punctuate the episode. A two-week ceasefire announced on 7 April 2026 committed to reopening the Strait, but collapsed within days when the Islamabad talks failed and the United States imposed a naval blockade on 13 April (International Monetary Fund 2026). A round of U.S. strikes followed on 25 May 2026, and the war was wound down only by a separate sixty-day ceasefire announced on 11 June 2026 (signed 17 June), which reopened the Strait. The April truce furnishes the within-crisis reversal the analysis exploits: under the safe-haven hypothesis of Section 3.3, a military/oil-supply shock should draw a haven bid into the dollar as the threat builds and release it on credible de-escalation, so that any premium the closure put into the dollar should come back out as the Strait is set to reopen. The June ceasefire falls too close to the end of the sample for a full event window (Section 6.5).

4.3 The Ukraine Control

The Russian invasion of Ukraine on 24 February 2022 serves as the external benchmark: a military/oil-supply shock the United States did not author and in which it was not a belligerent. The invasion combined a sharp spike in geopolitical risk with

a supply-driven jump in energy prices, both visible in Figures 4.2 and 4.3. Episodes of this kind are the empirical reference point for the classical safe-haven response reviewed in Section 2.3: in risk-off events the dollar, yen and Swiss franc tend to appreciate against other currencies.

Because the United States is here a bystander rather than a protagonist, the invasion isolates the *nature* of the shock from U.S. authorship. Under the hypothesis of Section 3.3 the dollar should display the classical haven response, making the 2022 invasion the benchmark against which the U.S.-authored tariff shock and the U.S.-involved Hormuz crisis are compared.

4.4 The Structure of the Comparison

Taken together, these three episodes populate the originator–nature taxonomy of Section 3.2 in a way that gives the comparison its identifying power. The invasion of Ukraine on 24 February 2022 is the external benchmark, a military/oil-supply shock the United States did not author; the Liberation Day tariffs of 2 April 2025 are the U.S.-authored trade-policy shock; and the Hormuz crisis, which began with the U.S.–Israeli airstrikes of 28 February 2026 and the closure of the Strait of Hormuz on 2 March, is a military/oil-supply shock in which the United States is implicated as a co-belligerent. Holding the crisis character of the shock roughly fixed while varying its origin is what allows the dollar’s response to be attributed to the *nature* of a shock rather than to U.S. authorship, or the reverse: if origin governs the response, the two U.S.-linked episodes should move the dollar alike and apart from Ukraine, whereas if nature governs it, the two military/oil-supply shocks should resemble each other regardless of who authored them.

Around each treatment sits a dated reversal that sharpens the reading. The ninety-day tariff pause of 9 April 2025 partially unwinds the Liberation Day shock, and the two-week ceasefire of 7 April 2026 unwinds the Hormuz threat; if a safe-haven premium moves the dollar on the shock, it should move back, at least in part, on its reversal. Two additional military/oil-supply episodes – the twelve-day Israel–Iran war that began on 13 June 2025 and the U.S. strikes of 25 May 2026 – provide further events against which the main results can be checked. Each of these dates is treated as the centre, t_0 , of the event window defined in Chapter 5, and announcement dates are used throughout, since these are the points at which the information reaches markets.

4.5 Market Backdrop, 2019–2026

Figures 4.1–4.3 place the episodes in their longer-run context, plotting the broad dollar index, the price of Brent crude, and the [Caldara and Iacoviello \(2022\)](#) geopolitical-risk index over 2019–2026, with the four main events marked on each series. These figures are intended to orient the reader to the level and volatility of each series around the events; the formal measurement of how the dollar responded, and of whether that response differed across episodes, is deferred to Chapter 6.



Figure 4.1: Broad dollar index, 2019–2026, with the four main events marked on the series: the Ukraine invasion (24 February 2022), Liberation Day (2 April 2025), the twelve-day war (13 June 2025), and the Hormuz crisis (28 February 2026). Source: FRED.



Figure 4.2: Brent crude oil price (USD/bbl), 2019–2026, with the same four events marked as in Figure 4.1. Source: FRED.

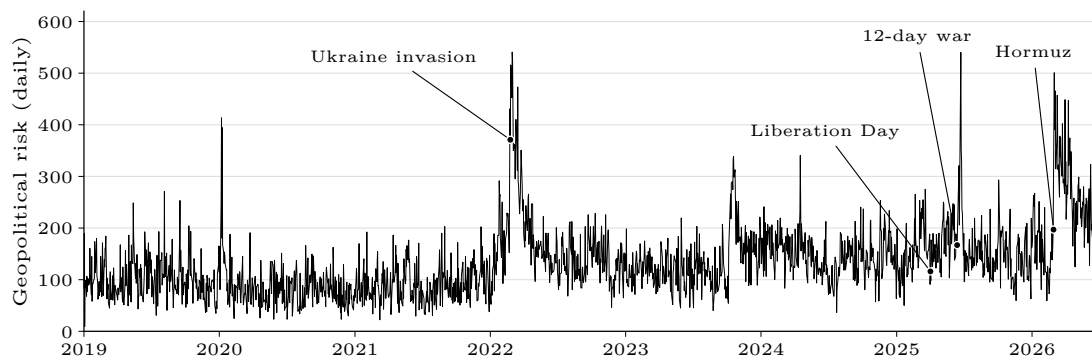


Figure 4.3: Caldara–Iacoviello daily Geopolitical Risk Index, 2019–2026, with the same four events marked as in Figure 4.1. Source: Caldara and Iacoviello’s Geopolitical Risk index ([Caldara and Iacoviello 2022](#)).

5 Data and Methodology

5.1 Data

5.1.1 Exchange Rates and Financial Variables

The currency panel comprises 32 exchange rates against the U.S. dollar, sampled daily from 1 January 2019 to 30 June 2026. Spot rates are taken from LSEG/Refinitiv at the WM/Reuters (WMR) fixing and exported as a single consistent panel; the Federal Reserve’s H.10 release (via FRED) serves as a cross-check. All rates are expressed as foreign currency per U.S. dollar, so that a rise denotes dollar appreciation; the four rates LSEG quotes in the opposite convention (EUR, GBP, AUD, NZD) are inverted accordingly. Daily returns are log changes, $r_{i,t} = \ln(S_{i,t-1}/S_{i,t})$, signed so that a positive value is an appreciation of the foreign currency against the dollar.

The macro-financial block is drawn from FRED: the broad, advanced-economy and emerging-market nominal dollar indices (DTWEXBGS, DTWEXAFEGS, DTWEXEMEGS); the 5- and 10-year Treasury yields (DGS5, DGS10), the 5-year TIPS yield (DFII5) and 5-year breakeven inflation (T5YIE); the VIX (VIXCLS); Brent and WTI crude (DCOILBRENTU, DCOILWTIC0); Henry Hub natural gas (DHHNGSP); and the S&P 500 (SP500). Three series not carried by FRED are taken from LSEG: the gold spot price (XAU=ZKBZ, USD/oz), the Euro Stoxx 50 price index (.STOXX50E, EUR) and the Dutch TTF front-month natural-gas contract (TRNLTTFMc1, EUR/MWh). Indices, equities and commodities enter as log returns; yields and the VIX enter in first differences.

5.1.2 Geopolitical and Policy Uncertainty Indices

Geopolitical risk is measured by the Caldara and Iacoviello (2022) Geopolitical Risk (GPR) index, used at daily frequency—with its *threat* and *act* sub-indices—and monthly, downloaded from the authors’ site. Policy uncertainty is captured by the U.S. daily Economic Policy Uncertainty (EPU) index and the Trade Policy Uncertainty (TPU) index from *policyuncertainty.com*. The dollar and carry risk factors are reconstructed from spot and one-month forward exchange rates following Lustig, Roussanov and Verdelhan (2011): the one-month forward discount $f_{i,t} - s_{i,t}$ proxies the interest-rate differential under covered interest parity; currencies are sorted on this signal each month, and the dollar factor (the average forward-implied excess return) and the carry factor (the high-minus-low portfolio return) are formed.

Rebuilding the factors from current market data extends them through the 2026 sample, replacing the course-supplied series that end in 2017.

5.1.3 Portfolio Construction

Two basket schemes are used. The first sorts currencies by their average one-month forward discount over the sample into terciles: the low-yield tercile forms the *safe* (funding) basket and the high-yield tercile the *risky* (carry) basket, with pegged and managed currencies (DKK, HKD, SAR, KWD, CNY) excluded from the sort. This is the classification that underlies the carry factor and the main event-study specification (Figure 5.1). Because the low-yield tercile also contains low-rate Asian currencies that are not conventional safe havens, a second, literature-standard scheme is reported as a robustness check: a *named-haven* safe basket of the yen, Swiss franc and euro against a *risky* basket of the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah. Independently, currencies are split by net oil-trade position into oil exporters (NOK, CAD, MXN, COP, BRL, SAR, KWD) and oil importers (JPY, KRW, INR, THB, TWD, TRY). All baskets are equally weighted; the equal-weighted average of all currency returns defines the dollar basket (USD_EW).

5.2 Event Study Methodology

The main specification is the *constant-mean return* model. For each series i the normal return is its average over an estimation window of $[-130, -11]$ trading days preceding the event; the abnormal return on day t of the event window is the deviation from that mean,

$$AR_{i,t} = R_{i,t} - \hat{\mu}_i, \quad \hat{\mu}_i = \frac{1}{120} \sum_{s=-130}^{-11} R_{i,s}, \quad (5.1)$$

and the cumulative abnormal return aggregates these over the window,

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{i,t}. \quad (5.2)$$

The event window is $[-20, +20]$ trading days. Cumulative windows $CAR(0, h)$ for $h \in \{1, 5, 10, 20\}$ capture the announcement reaction, while a long horizon $CAR(0, 50)$ —estimated on $[-170, -51]$ so the estimation window does not overlap the wider

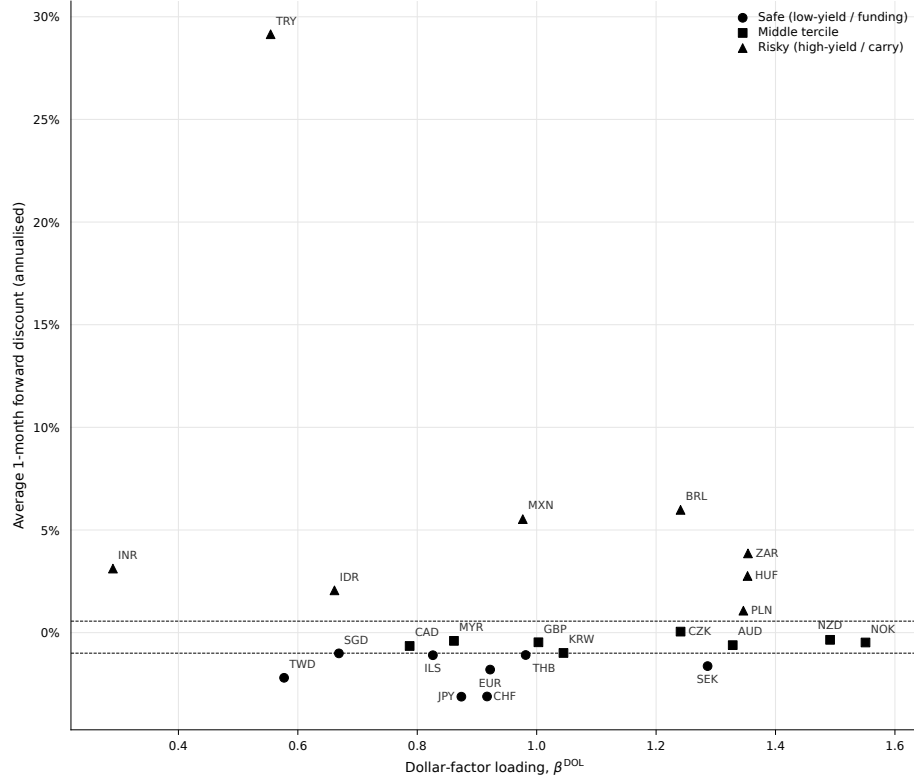


Figure 5.1: Currency classification. Each currency is plotted by its dollar-factor loading β^{DOL} (from a time-series regression of its monthly excess return on the reconstructed dollar and carry factors) against its average one-month forward discount, the signal used for the tercile sort. Dashed lines mark the tercile cut-offs, so marker shape (bucket) and vertical position agree: low forward discount $\rightarrow$ safe (funding), high $\rightarrow$ risky (carry).

event window—gauges persistence. The significance of a single CAR is assessed with $t = \text{CAR}/(\hat{\sigma}_i\sqrt{L})$, where $\hat{\sigma}_i$ is the estimation-window standard deviation of AR_i and $L = t_2 - t_1 + 1$. The difference between two events' CARs is tested with $\Delta = \text{CAR}_A - \text{CAR}_B$ and standard error $\sqrt{L(\hat{\sigma}_A^2 + \hat{\sigma}_B^2)}$, treating the (non-overlapping) event windows as independent.

The constant-mean model is preferred to the market model because the object of study is the dollar itself. The natural market factor for the currency baskets is the dollar factor, which is endogenous to the response under examination: regressing dollar exchange rates on a dollar factor would absorb the common dollar movement into $\beta_i R_{m,t}$ and leave little in the abnormal return. As a robustness check on *individual* currencies, the market model

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}, \quad \text{AR}_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}), \quad (5.3)$$

with $R_{m,t}$ the equal-weighted dollar factor, is reported in Appendix 10 and leaves the safe-haven hierarchy unchanged. Where cross-sectional dependence across currencies is a concern, the Kolari and Pynnönen (2010) adjusted test can replace the plain t -test; given the small number of non-overlapping events, that refinement is relegated to the appendix.

5.3 Regression Specification: The Safe-Haven Breakdown

The event study measures *whether* the dollar moved; a complementary regression measures *whether its relationship with risk changed*. The classical safe-haven property is that the dollar appreciates when global risk rises, so the daily broad-dollar return is regressed on the daily change in the VIX, interacted with a dummy for each crisis window:

$$r_t^{\text{USD}} = a + b_1 \Delta \text{VIX}_t + \sum_k \beta_k (\Delta \text{VIX}_t \times D_t^k) + \sum_k \gamma_k D_t^k + \varepsilon_t, \quad k \in \{\text{tariff, Hormuz, Ukraine}\}, \quad (5.4)$$

where D_t^k equals one from the event day through twenty trading days after shock k . A positive ΔVIX is a risk-off move and a positive r^{USD} a dollar appreciation, so $b_1 > 0$ is the normal safe-haven reflex; a negative interaction β_k means the reflex weakened during episode k , and the dollar's risk-sensitivity within that window is $b_1 + \beta_k$. Standard errors are Newey–West (HAC, five lags), robust to the volatility

5 Data and Methodology

clustering of daily returns. Results are reported in Section 6.6.

6 Empirical Results

This chapter presents the thesis’s two empirical layers. An event study (Sections 6.1–6.5) measures how the U.S. dollar, the cross-section of currency baskets, and crude oil responded to four shocks; a complementary regression (Section 6.6) then tests whether the dollar’s defining relationship with risk held or broke across the episodes. The two layers reinforce one another, reaching the same conclusion by different routes.

The event study isolates the thesis’s central contrast: three of the shocks are *U.S.-originated* (the Liberation Day tariff announcement, the twelve-day war, and the 2026 Hormuz crisis), while the 2022 Russian invasion of Ukraine is an *external* military/oil shock in which the United States is not the instigator, and which therefore serves as a benchmark for the “classical” safe-haven response. Abnormal returns are computed with a constant-mean model: the normal return of each series is its mean over the estimation window $[-130, -11]$ trading days, the abnormal return is the deviation from that mean, and the cumulative abnormal return $CAR(0, h)$ sums abnormal returns from the event day through trading day h .¹ Significance is assessed with a t -test on the CAR, $t = CAR/(\hat{\sigma}_{\text{est}}\sqrt{L})$. Tables 6.1 and 6.2 collect the headline numbers; the discussion below reads them event by event.

6.1 CARs: Tariff Shock (Liberation Day)

The April 2, 2025 tariff announcement triggered an immediate and *persistent* depreciation of the dollar. The broad dollar index fell 1.41% on the announcement day (significant at the 1% level) and continued to slide: -2.81% by ten trading days, -3.62% by twenty, and -6.34% by fifty (Table 6.1, Panel A). The 90-day tariff pause on April 9 did not arrest the move—if anything it deepened it, the broad index reaching -6.74% fifty days out—indicating that markets read the episode as a durable shift in the U.S. policy regime rather than a transient shock.

The cross-section confirms that the safe-haven premium did not accrue to the dollar but *leaked away from it*. Over the five-day window the named-haven basket—the yen, Swiss franc and euro—appreciated 2.02% against the dollar, the move concentrated

¹The constant-mean specification is preferred to the market model here because the object of study is the dollar itself: the natural “market” factor for the currency baskets is the dollar factor, which is endogenous to the response we seek to measure and would absorb the common dollar movement into $\beta R_{m,t}$. The market model with the dollar factor is therefore reported only as a robustness check on *individual* currencies (Appendix 10). See Section 5.2.

6 Empirical Results

Table 6.1: The U.S. dollar’s response by event and horizon

Event	Broad USD index — CAR(0, h), %				
	$h=1$	$h=5$	$h=10$	$h=20$	$h=50$
<i>Panel A. U.S. trade-policy shock</i>					
Liberation Day	−1.41***	−0.00	−2.81**	−3.62**	−6.34***
Tariff pause	−1.46***	−3.01***	−3.32***	−4.29***	−6.74***
<i>Panel B. U.S. military / oil-supply shock</i>					
Twelve-day war	+0.06	+0.94	−0.03	+0.55	−0.12
Hormuz crisis	+1.40***	+1.54***	+2.12***	+3.29***	−0.13
Hormuz ceasefire	−1.14***	−1.72***	−1.65**	−1.47	+0.17
<i>Panel C. External control (non-U.S.)</i>					
Ukraine invasion	+0.50	+1.06*	+1.70**	+0.94	+3.12**

Notes. Cumulative abnormal returns of the broad nominal U.S. dollar index (FRED DTWEXBGS), in percent, from a constant-mean model. Estimation window $[-130, -11]$ trading days for $h \leq 20$; $[-170, -51]$ for $h=50$ (pushed back so it does not overlap the wider event window). Positive values denote dollar appreciation. Tariff pause and Hormuz ceasefire are sub-events within their parent windows; “Hormuz ceasefire” is the two-week ceasefire of 7 April 2026, distinct from the later 60-day ceasefire of 11 June 2026. The US strikes (25 May 2026) and that 60-day June ceasefire are omitted because the sample ends 2026-06-16, leaving too few post-event days for a stable window. *, **, *** denote significance at the 10%, 5% and 1% level.

in the franc (+3.08%, significant at the 1% level); the broader carry-sorted safe basket, which also contains low-yield Asian currencies, gained a more muted 0.97% (Table 6.2). Either way the flight to quality went to competing havens rather than to the dollar. Crude oil fell sharply—Brent −14.24% and WTI −13.19% over five days—European equities sold off hard (Euro Stoxx 50 −14.58%), and the oil-exporter basket slipped 0.90%, consistent with the tariff shock operating through a global *demand*-destruction channel rather than a supply channel. This is the signature of the United States acting as a destabiliser: a U.S.-made shock that simultaneously weakens the dollar, depresses oil and equities, and redirects safe-haven flows to competing currencies.

6.2 CARs: Iran/Hormuz Shock

The two military/oil-supply episodes are reported separately because they differ in both intensity and the role the data assign to the United States.

Twelve-day war (June 2025). The response is muted and statistically insignificant throughout: the broad dollar index moves +0.06% on day one and +0.94% over five

6 Empirical Results

Table 6.2: Cross-section of five-day responses: the mirror image

Series, CAR(0, 5) (%)	U.S.-originated			Control
	Liberation Day	Hormuz	Twelve-day war	Ukraine
Broad USD index	−0.00	+1.54***	+0.94	+1.06*
Dollar (equal-weighted)	+0.26	+1.39**	+0.89	+1.04
Safe currencies (carry)	+0.97	−1.03	−1.08	−1.04*
Risky currencies (carry)	−0.84	−1.88***	−0.91	−2.34*
Oil exporters	−0.90	−0.71	−0.49	+0.25
Oil importers	+0.85	−1.10**	−1.19	−0.72
Gold	−1.56	−4.22	−1.60	+1.34
Euro Stoxx 50	−14.58***	−8.23***	−2.79	−6.03**
Brent crude	−14.24***	+27.83***	+11.21***	+13.63***
WTI crude	−13.19***	+34.71***	+10.34**	+14.26***
<i>Robustness: named safe-haven baskets</i>				
Safe (JPY, CHF, EUR)	+2.02*	−1.25	−1.45	−0.68
Risky (TRY, BRL, ZAR, MXN, COP, IDR)	−1.77*	−1.24	−0.37	+0.20

Notes. Five-day cumulative abnormal returns (%), constant-mean model, estimation window $[-130, -11]$. Liberation Day is the trade-policy shock; Hormuz and the twelve-day war are the military/oil-supply shocks; Ukraine is the non-U.S. control. For the currency baskets a positive value means the basket *appreciated against the dollar*; for the dollar indices, gold and equities positive means a price increase, and for crude positive means a price increase. *Safe/Risky (carry)* are the bottom and top terciles of the rebuilt forward-discount classification (Section 5.1.3); the robustness panel reports the literature-standard named-haven baskets. Gold's CARs are baseline-sensitive given its exceptional 2025–26 trend and are read for sign rather than magnitude (Section 6.4). *, **, *** denote significance at the 10%, 5% and 1% level.

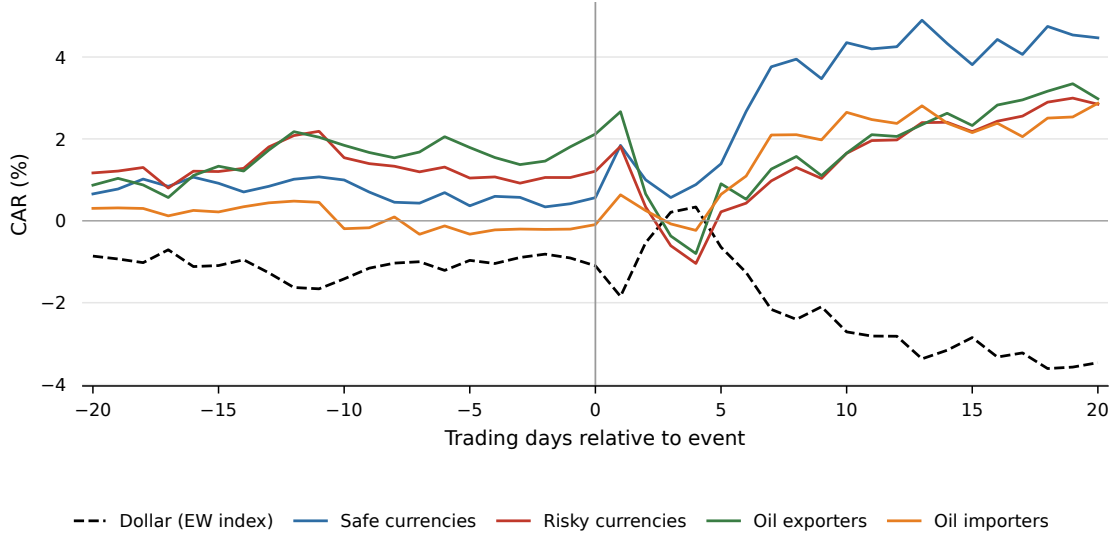


Figure 6.1: Liberation Day (2 April 2025): cumulative abnormal returns of the dollar and the currency baskets over ± 20 trading days. The dollar slides while the named-haven currencies bid; crude and European equities fall.

days, neither distinguishable from zero (Table 6.1, Panel B). Crude rose meaningfully (Brent +11.21% at five days), but the currency response is essentially flat. This is the “ambiguous actor” case anticipated in the introduction: with the United States a co-belligerent rather than a pure safe harbour, the flight-to-the-dollar reflex and the erosion pressure roughly offset, leaving no significant net move.

Hormuz crisis (February–March 2026). Here the dollar behaves as a classical safe haven. The broad index appreciates 1.40% on impact, building to +2.12% at ten days and +3.29% at twenty (Table 6.1). Crude spikes violently—Brent +27.83% and WTI +34.71% over five days—and the currency cross-section sorts on oil exposure: the oil-importer basket (yen, won, rupee) depreciates 1.10% against the surging dollar, underperforming the oil-exporter basket, which falls only 0.71% as higher crude cushions exporters. The dollar is bid even though the United States is the originator of the strikes, indicating that for a security/oil-supply shock the safe-haven reflex survives U.S. implication—in sharp contrast to the trade-policy shock.

Crucially, the Hormuz appreciation is *transitory*. By fifty days the broad-index CAR has returned to −0.13% (insignificant): the entire move unwinds. The clearest evidence that this was a genuine but temporary risk premium comes from the **April 7, 2026 ceasefire**, a within-crisis sub-event marked at +26 days in Figure 6.3. On the two-week ceasefire announcement—which committed to reopening the Strait of Hormuz—the dollar gave back 1.14% on the day and 1.72% over five days, and crude collapsed roughly 22% within ten days. The premium that the threat put *into* the dollar came back *out* on de-escalation, mapping the exchange-rate move directly onto the geopolitical risk it priced. This round-trip is about as close to clean identification as a single-country event study allows.

6.3 Cross-Event Comparison

Reading across Table 6.2, the two U.S.-originated shocks produce a *mirror image*. The trade-policy shock weakens the dollar and depresses oil; the military/oil-supply shock strengthens the dollar and lifts oil. The dollar is therefore not a fixed safe haven or a fixed casualty—its response is conditioned by the *type* of shock, even holding the originator (the United States) fixed.

This contrast is not merely descriptive. Table 6.3 tests the difference in the dollar’s CAR between events directly. At the ten-day horizon the broad dollar index is 4.93 percentage points lower under the tariff shock than under the Hormuz crisis

6 Empirical Results

($t = -3.7$) and 4.51 points lower than under Ukraine ($t = -3.4$), both significant at the 1% level; by twenty days the tariff–Hormuz gap widens to 6.9 points. Crucially, the Hormuz and Ukraine responses are statistically *indistinguishable* (differences of 0.4–2.4 points, none significant): the dollar appreciates after the U.S.-originated Hormuz shock by as much as it does after the external Ukraine shock. The mirror image is thus driven entirely by the trade-policy shock—the only episode that moves the dollar significantly differently from a classical safe-haven event. The divergence is a horizon phenomenon: at five days no pair is yet significant, consistent with the announcement reaction taking up to two weeks to resolve.

Table 6.3: Difference in dollar CARs across events (broad USD index, %)

Contrast (A – B)	$\Delta\text{CAR}(0, h)$, percentage points		
	$h=5$	$h=10$	$h=20$
Liberation Day – Hormuz crisis	–1.54	–4.93***	–6.92***
Liberation Day – Ukraine	–1.06	–4.51***	–4.56**
Hormuz crisis – Ukraine	+0.48	+0.42	+2.35
Liberation Day – Twelve-day war	–0.94	–2.78*	–4.17*

Notes. Difference between two events’ cumulative abnormal returns of the broad nominal U.S. dollar index (%), constant-mean model, estimation window $[-130, -11]$. Standard error $\sqrt{L(\sigma_A^2 + \sigma_B^2)}$ with $L = h + 1$; events treated as independent (windows do not overlap). A positive value means the dollar rose more (or fell less) under shock A than shock B. */**/***: 10/5/1%.

The Ukraine benchmark disciplines the interpretation. As an external military/oil shock, it generates the textbook response—the dollar appreciates (+1.70% at ten days, +3.12% at fifty) and crude rises 13.63% (Table 6.2)—and, as just shown, that response is statistically the same as Hormuz. Two features nonetheless distinguish the external from the U.S.-originated military shock. First, the Ukraine appreciation is durable whereas the Hormuz appreciation reverts, consistent with an external shock conferring a lasting safe-haven bid and a U.S.-implicated one only a temporary one. Second, and most importantly for the thesis, the *trade-policy* shock breaks the pattern entirely: the only episode in which the dollar depreciates is the one the United States itself authored on the economic-policy front. The safe-haven function is intact for geopolitical/oil-supply shocks—even U.S.-originated ones—but it inverts for a U.S. trade-policy shock.

6 Empirical Results

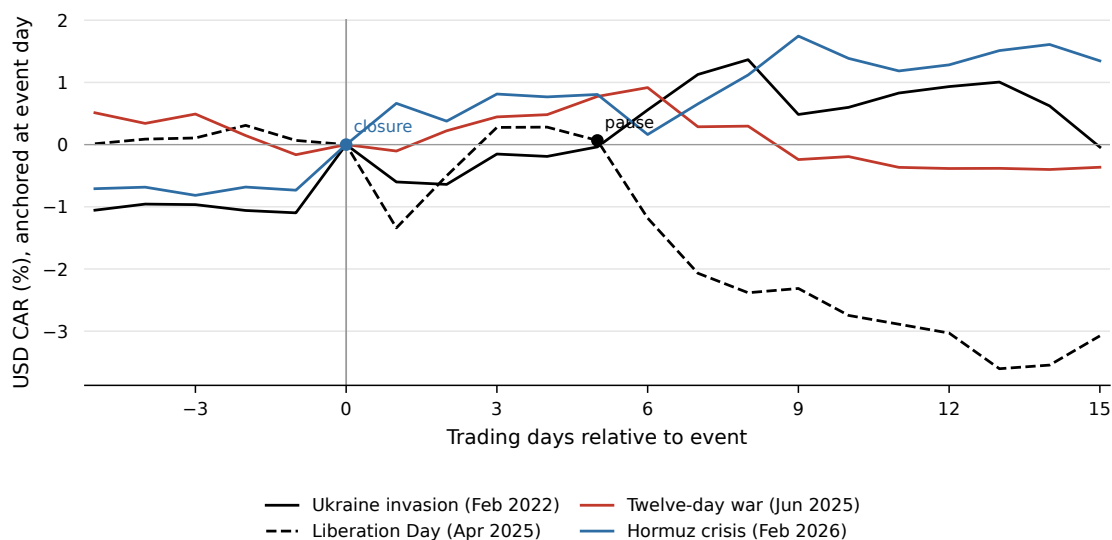


Figure 6.2: Broad dollar CARs across the four headline events, re-anchored on the event day (-5 to +15). The tariff shock is the only episode in which the dollar depreciates; Hormuz and Ukraine trace the classical appreciation.

6.4 Gold and Commodities

The commodity dimension reinforces the demand-versus-supply reading. Brent and WTI move in opposite directions across the two shock types—down $\sim 14\%$ on the tariff shock (a demand shock) and up $\sim 28\text{--}35\%$ on the Hormuz supply shock—and the two benchmarks agree closely, so the result does not depend on the choice of crude reference (Section 6.5 returns to this). The oil response is the channel through which the cross-section of oil-exporter and oil-importer currencies sorts in Table 6.2.

Gold tells a subtler story than the “universal haven” a first intuition would suggest. Rather than rising around every shock, gold tracks the *inverse of the dollar* across the U.S.-originated episodes: it gains when the tariff shock weakens the dollar ($+5.99\%$ at ten days, and $+8.48\%$ around the tariff pause) and falls when the Hormuz shock strengthens it (-8.36% at ten days, deepening at longer horizons). The one departure is the external control: around the Ukraine invasion gold and the dollar rise *together* ($+4.24\%$ for gold at ten days), the dual-haven response a classical risk-off episode produces. Gold is thus not an unconditional substitute for the dollar but a hedge against it—for the U.S.-made shocks it moves opposite the dollar, reinforcing that the dollar’s own conditional response, and not a generic flight to safety, is the organising variable of this chapter; only for the external shock do both havens bid at once. Two caveats temper the magnitudes: gold appreciated roughly 240% over the sample, so the constant-mean baseline is unusually sensitive and the long-horizon gold CARs

(e.g. Hormuz at twenty days) overstate the move—the *sign* is the robust feature, not the level—and the single-contributor quote used for the gold fix (LSEG XAU=ZKBZ) is noted in Appendix 10.

Equities corroborate the risk-off reading without the dollar’s conditionality. The Euro Stoxx 50 falls around every shock—−14.58% on Liberation Day, −8.23% around Hormuz, and −6.03% after the Ukraine invasion (all significant at 5% or better)—so European risk assets sell off whether the shock is a trade war, an oil-supply scare, or an external invasion. The contrast with the dollar is the point: equities are an *unconditional* casualty of geopolitical and policy shocks, whereas the dollar’s sign flips with the type of shock. The safe-haven question is therefore specific to the dollar, not a generic property of risk assets.

6.5 Robustness

Three checks support the readings above. First, *longer windows* confirm the persistence pattern (Table 6.1, $h=50$): the tariff depreciation deepens to −6.34%, the Hormuz appreciation fully reverts, and the Ukraine appreciation persists at +3.12%—the long horizon thus separates a level re-rating of the dollar (tariffs) from a transitory risk premium (Hormuz). The ± 50 -day figures (e.g. Figure 6.3) plot crude on a secondary axis and mark neighbouring sub-events, making both the persistence and the ceasefire reversal visible. Second, *Brent versus WTI*: the crude responses are quantitatively similar across all events (Table 6.2), so conclusions are invariant to the benchmark. Third, the *market model* with the dollar factor, reported for individual currencies in Appendix 10, leaves the safe-haven hierarchy unchanged.

Two limitations bound the long-horizon results. The ± 50 -day windows of the two 2025 events overlap (Liberation Day’s tail reaches the twelve-day war ~ 52 trading days later), so their long-horizon CARs are not fully independent; this is why the persistence figures are read event by event, while the formal cross-event *difference* tests (Table 6.3) use horizons up to twenty days, at which the windows of the compared pairs do not overlap. And the two most recent shocks—the May 2026 U.S. strikes and the June 2026 ceasefire—are omitted from the longer horizons because the sample ends 2026-06-16, leaving too few post-event trading days for a stable window.

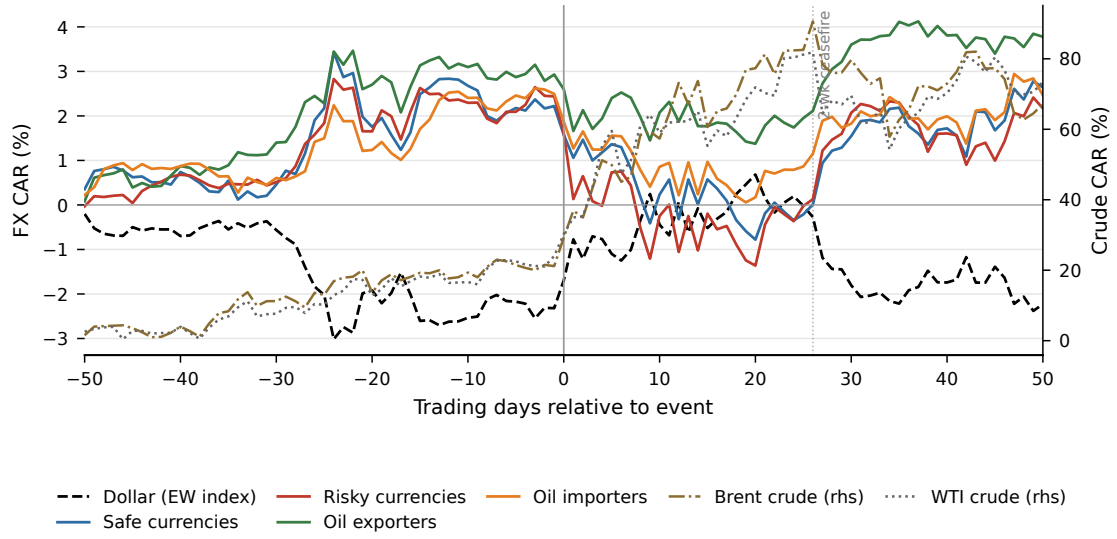


Figure 6.3: Hormuz crisis, ± 50 trading days. Left axis: currency-basket CARs; right axis: Brent and WTI CARs. The dollar’s appreciation reverts by +50; the April 7 two-week ceasefire (marked) both crashes crude and unwinds the dollar premium.

6.6 Regression Evidence on the Safe-Haven Breakdown

The event study above shows *that* the dollar moved in opposite directions across the two shock types. This section asks the sharper question directly: did the dollar’s defining safe-haven property—appreciating when global risk rises—actually hold or break in each episode? The dummy-interaction regression specified in Section 5.3 answers it with ordinary least squares and heteroskedasticity-robust standard errors, requiring none of the modelling assumptions of a conditional-correlation (GARCH) approach.

The result is unambiguous (Table 6.4). On normal days the dollar is a safe haven: the coefficient on the change in the VIX is positive and strongly significant ($b_1 = 0.0003$, $t = 7.0$), so a risk-off jump in volatility lifts the dollar. During the **tariff** window this reflex *disappears*: the interaction is equal and opposite ($\beta_{\text{tariff}} = -0.0003$, significant at the 1% level), so the within-window sensitivity $b_1 + \beta_{\text{tariff}}$ is essentially zero. The dollar stopped behaving as a haven precisely when the United States was the source of the shock. During the **Hormuz** window the interaction is small and statistically insignificant ($t = 1.3$), leaving the safe-haven relationship unchanged from normal; during the **Ukraine** window it is, if anything, *stronger* ($\beta = +0.0008$,

6 Empirical Results

Table 6.4: Did the dollar’s safe-haven reflex survive each shock? (OLS, HAC standard errors)

	Coef.	t	Within-window slope $b_1 + \beta_k$
ΔVIX (normal, b_1)	+0.0003***	7.0	—
$\times$ Tariff (β_{tariff})	−0.0003***	−2.6	≈ 0.0000
$\times$ Hormuz (β_{Hormuz})	+0.0003	1.3	+0.0006
$\times$ Ukraine (β_{Ukraine})	+0.0008***	4.8	+0.0011
<i>Window (level) dummies</i>			
Tariff window	−0.0015*	−1.7	
Hormuz window	+0.0011**	2.1	
Ukraine window	+0.0010**	2.1	

Notes. OLS of the daily broad-dollar return on the daily change in the VIX, its interactions with crisis-window dummies, and the window dummies; Newey–West (HAC, 5 lags) standard errors. Crisis windows run from the event day to +20 trading days. $N = 1,944$, $R^2 = 0.06$. A positive coefficient on ΔVIX means the dollar appreciates when risk rises (safe-haven behaviour); the near-zero within-window slope for the tariff episode is the breakdown. */**/***: 10/5/1%.

$t = 4.8$). The window (level) dummies echo the event study—the average dollar drift is negative in the tariff window and positive in the Hormuz and Ukraine windows.

This regression reaches the event study’s conclusion by a different and fully transparent route: the dollar’s safe-haven function broke down for the U.S. trade-policy shock but remained intact for the geopolitical and oil-supply shocks. Because it rests on a single, interpretable least-squares specification with robust standard errors, it is the test most directly tied to the central hypothesis—and it delivers that test at a fraction of the modelling and estimation risk of a time-varying-volatility alternative.

7 Discussion

This chapter draws the event-study and regression results together, relates them to the literature, and sets out their implications and limits.

7.1 Findings in Light of the Literature

The results speak directly to the five literatures set out in Chapter 2. The sharpest verdict is for the *dollar-erosion* strand. [Kamin \(2025\)](#) and [Jiang et al. \(2026\)](#) read the 2025 tariff episode as evidence that the dollar has stopped behaving as a safe haven, and [Hassan et al. \(2025\)](#) supply the theory: a trade war that isolates U.S. goods markets erodes the currency’s safety premium. The findings here both confirm and bound that claim. They confirm it for the tariff shock—the dollar depreciates and its risk sensitivity collapses to zero—but they bound it by showing that the erosion is not a general weakening: the very same dollar appreciates around the contemporaneous Hormuz shock. What the dollar-erosion papers identify as a regime change is, on this evidence, specific to U.S.-originated economic-policy shocks rather than a property of the post-2025 dollar as such.

For the *safe-haven* tradition ([Ranaldo and Söderlind 2010](#); [Habib and Stracca 2012](#); [Bock and Carvalho Filho 2015](#)) the evidence is corroborative. The dollar performs as that literature predicts around both military/oil-supply shocks, appreciating as risk rises, and the formal difference test cannot separate the U.S.-implicated Hormuz response from the external Ukraine benchmark—so the haven bid survives even U.S. involvement when the shock is a security one. The conditional, crisis-contingent definition of [Baur and Lucey \(2010\)](#) is thus vindicated for the dollar: its haven status is real but state-dependent. That same framing organises the *gold* result. Gold hedges against the dollar rather than serving as a universal safe haven—it bids when the tariff shock weakens the dollar and falls when the Hormuz shock strengthens it, rising alongside the dollar only for the external Ukraine shock—exactly the state-dependence Baur and Lucey’s distinction implies.

The *commodity-currency* literature ([Y.-C. Chen and Rogoff 2003](#); [Kilian 2009](#)) is borne out selectively: oil-exporter currencies outperform importers only around the supply-driven Hormuz shock, the asymmetry Kilian’s typology implies, while the demand-driven tariff decline in oil produces no comparable sorting. Finally, the *geopolitical-risk* literature ([Caldara and Iacoviello 2022](#)) supplies both the measurement and a mechanism—the within-crisis ceasefire reversal, which unwinds the dollar

premium as the threat recedes, is the threat-versus-realisation distinction Caldara and Iacoviello emphasise, mapped onto the exchange rate.

7.2 The Dollar’s Safe Haven Is Conditional on Shock Type

The central finding is that the dollar’s safe-haven response is conditioned by the *type* of shock, not merely by who causes it. The dollar appreciated around both military/oil-supply shocks—the external Ukraine invasion and the U.S.-originated Hormuz crisis—and the formal difference test cannot distinguish the two responses, even though the United States authored one and merely witnessed the other. Yet it depreciated around the one purely economic-policy shock it created, the Liberation Day tariffs. Origin alone does not explain the pattern; the nature of the shock does. This refines the “dollar-erosion” reading ([Ostry et al. 2025](#); [Kamin 2025](#)): the safe-haven function is not weakening in general but is specifically suspended when the United States is itself the source of an economic-policy shock, while it survives intact—reversion notwithstanding—for the security shocks that the safe-haven tradition has always emphasised ([Ranaldo and Söderlind 2010](#); [Habib and Stracca 2012](#)).

The regression sharpens this further. Estimated over the full sample, the dollar’s sensitivity to risk is significantly positive on normal days—the textbook safe-haven reflex—but the interaction term shows it collapsing to essentially zero inside the tariff window, while it remains intact during the Hormuz and Ukraine windows. The breakdown is thus not merely a difference in the *level* of the dollar across episodes but a change in its very *relationship with risk*: during the U.S. trade-policy shock the dollar stopped trading like a safe haven, and only then.

7.3 Gold and the Oil Channel

Gold does not behave as the universal safe haven that a first reading of [Baur and Lucey \(2010\)](#) would predict. Instead it tracks the inverse of the dollar across the U.S.-originated shocks—rising when the tariff shock weakened the dollar, falling when the Hormuz shock strengthened it—and bids alongside the dollar only for the external Ukraine shock. Gold is thus better described as a hedge *against* the dollar than as a substitute for it, which reinforces, rather than competes with, the chapter’s

main message: the dollar’s own conditional move is the organising variable, and gold prices it from the other side.

The oil channel operates as the commodity-currency literature predicts, but only for the supply shock. During the Hormuz crisis, oil-importer currencies underperformed oil-exporters as crude surged, consistent with [Y.-C. Chen and Rogoff \(2003\)](#) and [Kilian \(2009\)](#); during the demand-driven tariff decline the sorting was muted. The yen illustrates the resulting tension: a classical safe haven that is also a large oil importer, it is pulled in opposite directions during an oil-supply shock, which is part of why the basket responses around Hormuz are smaller and noisier than the dollar-index response.

7.4 Policy Implications

Three implications follow. For reserve managers and central banks, the dollar’s safe-haven status cannot be treated as unconditional: hedging programmes calibrated on the assumption that the dollar always rallies in risk-off episodes will misfire precisely when the shock originates in U.S. economic policy. For portfolio managers, diversification should be thought of across *shock types*, not only across assets—gold and competing-haven currencies, not the dollar, carried the flight to quality in the tariff episode. For the broader debate on the dollar’s reserve role ([Obstfeld and Zhou 2023](#); [Jiang et al. 2026](#)), the evidence is a caution against over-reading any single episode: the 2025 tariff shock did dent the dollar, but the 2026 security shock showed the haven bid still firmly in place.

7.5 Limitations

Several caveats bound the results. The two most recent events—the May 2026 U.S. strikes and the June 2026 ceasefire—have too few post-event trading days for the longer windows, so they are reported only at short horizons. The regression’s crisis windows are defined around the events rather than by a structural break test, and the VIX–dollar link it measures is a reduced-form association, not a causal channel. Gold’s exceptional 2025–26 trend makes its constant-mean CARs sensitive in level, so they are read for sign rather than magnitude. As a single-country event study around a small number of recent, partly overlapping episodes, the design trades breadth for the clean U.S.-versus-external contrast; and without forward/basis data beyond the one-month tenor, covered-interest-parity deviations are left for future

work.

8 Conclusion

This thesis asked whether the U.S. dollar’s safe-haven response depends on the type of shock that triggers a crisis, even when the United States is implicated in the shock itself. Two recent episodes provided an unusually clean test: the 2025 Liberation Day tariffs, a purely U.S. economic-policy shock, and the 2026 Iran/Hormuz crisis, a geopolitical oil-supply shock, with the 2022 Russian invasion of Ukraine as an external control.

The evidence gives a clear answer. The two shock types produced a mirror image. The dollar depreciated around the tariff shock—over three and a half percent on the broad index within twenty trading days—while it appreciated around the Hormuz crisis and around Ukraine, and a formal difference test confirms the tariff response is significantly distinct from both security episodes, which are themselves statistically indistinguishable from one another. The dollar’s haven status is therefore conditional on the *nature* of the shock, not on its origin: it survives a U.S.-involved security shock but inverts for a U.S.-involved economic-policy shock. A complementary regression confirms the breakdown statistically: the dollar’s safe-haven sensitivity to risk—significantly positive on normal days—collapses to zero in the tariff window, while holding through the Hormuz and Ukraine episodes. Two further results round out the picture: gold acts as a hedge against the dollar rather than as a universal safe haven, and the oil channel sorts currencies only for the supply-driven shock.

The thesis contributes the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets, carries a gold and oil benchmark through the same design to connect the safe-haven and commodity-currency literatures, and extends standard cross-sectional currency tools to mid-2026 on a consistent WMR dataset.

Several extensions suggest themselves. As the geopolitical situation continues to evolve, the sample can be extended so that the most recent episodes acquire full post-event windows. Richer forward and basis data would allow covered-interest-parity deviations to be brought into the analysis, and intraday data would tighten the identification around the announcement windows. A dynamic conditional-correlation (DCC-GARCH) model of the dollar–VIX relationship would test the safe-haven-breakdown hypothesis directly and is the natural next layer. Finally, a formal model of shock-dependent safe-haven demand—in which the currency of the shock’s originator loses its haven premium for policy shocks but not for security shocks—would give the empirical pattern documented here a theoretical foundation.

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10 Data Sources

All data are daily, 1 January 2019 to 30 June 2026. The macro-financial block (dollar indices, Treasury and TIPS yields, breakeven inflation, the VIX, Brent and WTI crude, Henry Hub gas, and the S&P 500) is from FRED. The 32-currency spot and one-month forward panel, gold, the Euro Stoxx 50, and the TTF gas contract are exported from LSEG at the WM/Reuters fixing. The Geopolitical Risk index is from Caldara and Iacoviello, and the Economic and Trade Policy Uncertainty indices from *policyuncertainty.com*. The analysis pipeline (`consolidate_lseg.py` → `get_data.py` → `build_fx_factors.py` → `02_build_returns.py` → the event-study scripts 03–05 and the safe-haven regression `07_safehaven_regression.py`) reproduces every table and figure in the thesis from these sources.

Appendix

1 Currency Universe and Classification

Table 1 lists the 32 currencies in the panel with their carry-sorted bucket (low/middle/high forward-discount tercile, pegged and managed currencies excluded from the sort) and their oil-trade role. The named safe-haven robustness baskets used in Chapter 6 are the yen, Swiss franc and euro (safe) against the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah (risky).

2 Robustness

Market model. Alongside the constant-mean specification, the event study computes market-model abnormal returns for the individual currencies, using the equal-weighted dollar factor as the market return (`Code/03_event_study.py`, method `market_model`). These are reported in the replication output; the safe-haven ordering across currencies is unchanged relative to the constant-mean results in Chapter 6.

Regression robustness. The safe-haven regression of Section 6.6 uses Newey–West (HAC) standard errors, so its inference is robust to the autocorrelation and volatility clustering of daily returns—the main reason a simple least-squares interaction is preferred here to a conditional-correlation model. The crisis windows are the same event-day-to-+20 windows used in the event study, so the regression and the event study rest on a common, transparent definition of each episode.

Table 1: Currency classification

Code	Currency	Carry bucket	Oil role
AUD	Australian dollar	middle	—
BRL	Brazilian real	high (risky)	exporter
CAD	Canadian dollar	middle	exporter
CHF	Swiss franc	low (safe)	—
CLP	Chilean peso	no 1M forward	—
CNY	Chinese yuan	pegged/managed	—
COP	Colombian peso	no 1M forward	exporter
CZK	Czech koruna	middle	—
DKK	Danish krone	pegged (EUR)	—
EUR	Euro	low (safe)	—
GBP	Pound sterling	middle	—
HKD	Hong Kong dollar	pegged (USD)	—
HUF	Hungarian forint	high (risky)	—
IDR	Indonesian rupiah	high (risky)	—
ILS	Israeli shekel	low (safe)	—
INR	Indian rupee	high (risky)	importer
JPY	Japanese yen	low (safe)	importer
KRW	South Korean won	middle	importer
KWD	Kuwaiti dinar	pegged	exporter
MXN	Mexican peso	high (risky)	exporter
MYR	Malaysian ringgit	middle	—
NOK	Norwegian krone	middle	exporter
NZD	New Zealand dollar	middle	—
PEN	Peruvian sol	no 1M forward	—
PLN	Polish zloty	high (risky)	—
SAR	Saudi riyal	pegged	exporter
SEK	Swedish krona	low (safe)	—
SGD	Singapore dollar	low (safe)	—
THB	Thai baht	low (safe)	importer
TRY	Turkish lira	high (risky)	importer
TWD	Taiwan dollar	low (safe)	importer
ZAR	South African rand	high (risky)	—

Notes. Carry buckets are terciles of the average one-month forward discount (Section 5.1.3); pegged/managed currencies and the three currencies without a clean one-month forward (CLP, COP, PEN) are excluded from the carry sort. Oil role follows the net oil-trade classification.

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